

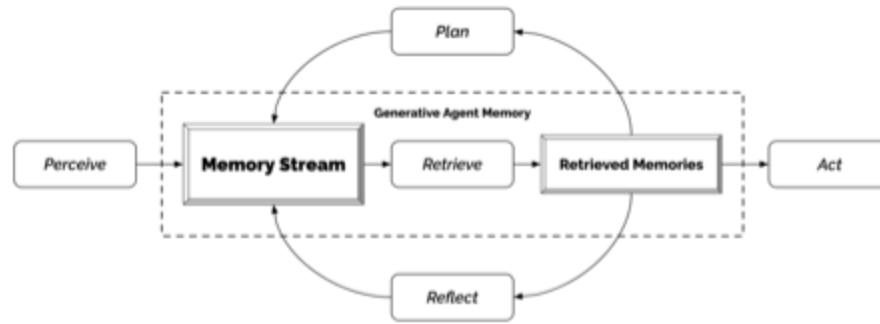
Working Note for Generative AI Safety

Silicon Dummy Doll: Addressing Interactive Risk of AI Usage Through Generative AI Agent.

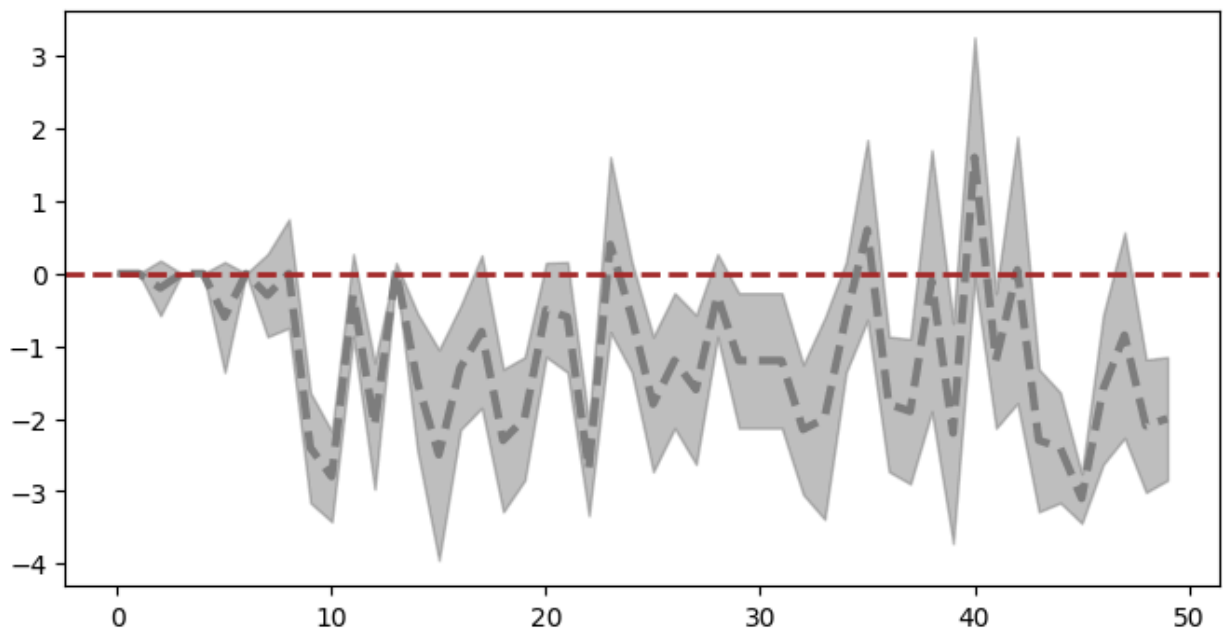
Large Language Models (LLMs) have demonstrated unprecedented capabilities in influencing human cognition across perceptual (Breum et al., 2024; Glickman & Sharot, 2025), emotional (Glickman & Sharot, 2025), social (Costello & Rand, 2024; Spitale et al., 2023), and political domains (Costello & Rand, 2024; Hackenburg et al., 2025). Recent studies suggest that these persuasive effects are not simply byproducts of language fluency or coherence, but are closely tied to LLMs' emergent internal value systems—systems that can subtly shape human judgment and behavior in ways that may diverge from established societal norms (Mazeika et al., 2025). This raises significant concerns: LLMs may inadvertently—or in some cases deliberately—promote perspectives or behaviors that are misaligned with human values, thereby influencing users at a psychological or ideological level.

While the risks posed by such influence have attracted growing scholarly attention (Weidinger et al., 2022; Röttger et al., 2025), most existing evaluations focus on model behavior in isolated, single-turn interactions (Ibrahim et al., 2024). However, real-world applications of LLMs increasingly involve extended, multi-turn dialogues that mirror natural conversations. In these dynamic settings, users are not passive recipients but active participants in an evolving social exchange. This shift underscores the need for evaluation frameworks that account for the contextual, temporal, and interactive dimensions of human-LLM engagement. Multi-turn interactions enable models to subtly adapt, reinforce, or steer user perspectives over time—effects that are difficult to capture with single-shot testing.

To address this methodological gap, we introduce *Silicon Dummy Doll*, a user simulation system built on a generative AI agent framework (Park et al., 2023). Our system enables researchers to simulate realistic multi-turn conversations under predefined scenarios and participation strategies, offering a scalable and automated way to study the longitudinal impacts of LLMs. Critically, *Silicon Dummy Doll* is designed not only to replicate user interaction patterns but also to surface interpretable indicators of model influence—tracing how LLM outputs may affect simulated users' beliefs, attitudes, or decisions. This feature makes it particularly valuable for stakeholders concerned with AI safety, ethics, and governance. Furthermore, the system's lightweight design ensures that it remains accessible to mid-sized research teams and institutions, making rigorous LLM evaluation both feasible and affordable.



[Prototype](#)



[Memory Sample](#)

- Related Work:

Ibrahim, L., Akbulut, C., Elasmr, R., Rastogi, C., Kahng, M., Morris, M. R., ... & Weidinger, L. (2025). Multi-turn Evaluation of Anthropomorphic Behaviours in Large Language Models. *arXiv preprint arXiv:2502.07077*.

Jin, Z., Heil, N., Liu, J., Dhuliawala, S., Qi, Y., Schölkopf, B., ... & Sachan, M. (2024, November). Implicit Personalization in Language Models: A Systematic Study. In *Findings of the Association for Computational Linguistics: EMNLP 2024* (pp. 12309-12325).

- Why it works:
 Lee, S., Peng, T. Q., Goldberg, M. H., Rosenthal, S. A., Kotcher, J. E., Maibach, E. W., & Leiserowitz, A. (2024). Can large language models estimate public opinion about global warming? An empirical assessment of algorithmic fidelity and bias. *PLOS Climate*, 3(8), e0000429.
 Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3), 337-351.
- How it connected to team building:
 The ethics/social science team provides research focus (what could be affected), scenario design (how that risk can occur), and interpretation (why that impact is possible).
- How it connected to risk prevention:
 More previous studies.
 A companion system alerting AI impacts
 A simple way: collect human-AI chat logs and target variables (e.g., partisanship before and after the experiments). Verify the effectiveness of the companion system.
 Build UI to alarm users when the risks are detected.
- Submission Plan:
[CHI 2026](#)
 Deadline: September 4th 2025
- Next week:
 Go through the related work.

References:

- Breum, S. M., Egdal, D. V., Mortensen, V. G., Møller, A. G., & Aiello, L. M. (2024, May). The persuasive power of large language models. In *Proceedings of the International AAAI Conference on Web and Social Media* (Vol. 18, pp. 152-163).
- Costello, T. H., Pennycook, G., & Rand, D. G. (2024). Durably reducing conspiracy beliefs through dialogues with AI. *Science*, 385(6714), eadq1814.
- Glickman, M., & Sharot, T. (2025). How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9(2), 345-359.
- Hackenburg, K., Tappin, B. M., Röttger, P., Hale, S. A., Bright, J., & Margetts, H. (2025). Scaling language model size yields diminishing returns for single-message political persuasion. *Proceedings of the National Academy of Sciences*, 122(10), e2413443122.

- Ibrahim, L., Huang, S., Ahmad, L., & Anderljung, M. (2024). Beyond static AI evaluations: advancing human interaction evaluations for LLM harms and risks. *arXiv preprint arXiv:2405.10632*.
- Mazeika, M., Yin, X., Tamirisa, R., Lim, J., Lee, B. W., Ren, R., ... & Hendrycks, D. (2025). Utility Engineering: Analyzing and Controlling Emergent Value Systems in AIs. *arXiv preprint arXiv:2502.08640*.
- Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023, October). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology* (pp. 1-22).
- Röttger, P., Pernisi, F., Vidgen, B., & Hovy, D. (2025, April). Safetyprompts: a systematic review of open datasets for evaluating and improving large language model safety. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 39, No. 26, pp. 27617-27627).
- Spitale, G., Biller-Andorno, N., & Germani, F. (2023). AI model GPT-3 (dis) informs us better than humans. *Science Advances*, 9(26), eadh1850.
- Weidinger, L., Uesato, J., Rauh, M., Griffin, C., Huang, P. S., Mellor, J., ... & Gabriel, I. (2022, June). Taxonomy of risks posed by language models. In *Proceedings of the 2022 ACM conference on fairness, accountability, and transparency* (pp. 214-229).